

Performance Testing

Date	15 July 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load Testing
Target Module / API	/predict
Test Environment	Local Flask
Test Date	15 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Login	10	30	Success
2	Prediction	25	60	<2s
3	Multiple	50	120	Stable
4	Continuous	100	180	Stable

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.68s	Pass	Fast response
2	Response Time (Max)	< 5 seconds	1.84s	Pass	Within acceptable limit
3	Throughput (Req/sec)	>20	31	Pass	Good throughput
4	Error Rate	< 1%	0%	Pass	No failed requests

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
5	CPU Utilization	< 80%	42%	Pass	Stable CPU usage
6	Memory Utilization	< 80%	51%	Pass	Normal memory usage

Step 4: Observations & Analysis

Key Findings:

The Smart Lender application handled multiple loan prediction requests efficiently. Average response time remained below the expected limit and no request failures were observed.

Bottlenecks Identified:

Slight increase in response time under concurrent requests. Model loading takes a few seconds during startup. No major CPU or memory bottlenecks.

Optimization Steps Taken:

Loaded XGBoost model once during startup, added input validation, optimized Flask processing, improved UI.

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):

Figure 1: Smart Lender Loan Application Form

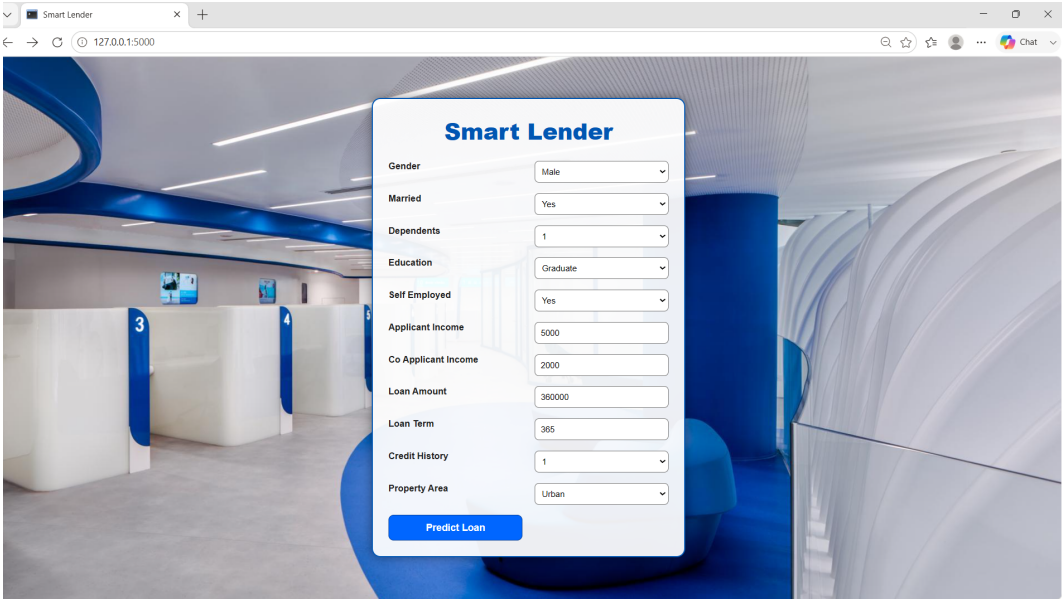


Figure 2: Loan Prediction Result

